
Residual-Aware Uncertainty Sampling Accelerates Autonomous Reconstruction of Heterogeneous Response Surfaces

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Abstract

Autonomous experiments that reconstruct an unknown response surface $\rho(\mathbf{p})$ over a multi-dimensional parameter space must decide, after every measurement, where to measure next so that the full map is recovered accurately with the fewest samples. Kriging-based decision engines steer these measurements with an acquisition function computed from the surrogate’s posterior. The reference methods for this setting are SMART, which samples the point of maximum posterior variance, and GSMART, which additionally weights that variance by the squared gradient of the surrogate to target sharp features. Both inherit the central limitation of a single global variogram: they cannot represent *local* structure, so they either sample roughly uniformly or chase a single local feature statistic. We introduce RAVE (Residual-Augmented Variance Exploration), an acquisition criterion that multiplies the posterior variance by a dimensionless, kernel-smoothed leave-one-out (LOO) residual field fitted to the very same correlation length as the surrogate. The residual field measures where the stationary surrogate fails to predict its own training data—a direct, model-consistent estimate of local reconstruction difficulty—and requires no additional hyperparameters beyond the length scale the experiment already fits. Across the five synthetic test functions of Noack et al. (2019)—two piecewise-sinusoidal phase-boundary maps, a smooth oscillatory surface, a non-differentiable piecewise surface, and a three-dimensional Stokes–Einstein diffusion map—RAVE reaches SMART’s maximum-budget error with roughly 32–40% fewer measurements on the three feature-bearing two-dimensional functions and at half the budget on the three-dimensional map, and is never worse than the baselines by more than statistical noise. We characterize the mechanism (residual steering concentrates samples where the stationary model is locally inadequate), the failure modes (it reduces to SMART when the response is statistically stationary, and a global variogram collapses under concentrated sampling unless stabilized), and the computational cost, which is comparable to the baselines.

1 Introduction

Modern scientific instruments can acquire data far faster than a human can decide what to measure next. In synchrotron X-ray scattering, for example, a focused beam probes a sample at discrete positions $\mathbf{p} = (x, y)$, and at each position a scalar quantity $\rho(\mathbf{p})$ —a scattering-peak intensity, a lattice parameter, or a structural order parameter—is extracted from the recorded diffraction pattern. The experimenter’s goal is frequently not to locate a single optimum but to *reconstruct the entire map* $\rho(\mathbf{p})$ over the sample. Because each measurement takes seconds to minutes, a full raster scan can consume hours of beamtime, so there is strong incentive to choose each subsequent measurement location intelligently Noack et al. [2019], Kusne et al. [2014], Lookman et al. [2016].

This problem—faithful recovery of an unknown function from as few evaluations as possible—is a central topic in the design and analysis of computer experiments Sacks et al. [1989], Santner et al. [2003], Stein [1999], Cressie [1990] and in the broader active-learning literature Settles [2012], MacKay [1992], Cohn et al. [1996], Houlsby et al. [2011]. A standard framework is a Gaussian process (GP) surrogate whose posterior variance defines an uncertainty field; an acquisition function is then maximized to choose the next point. The SMART framework of Noack et al. [2019] is a Kriging instantiation of this idea, using an exponential variogram $\gamma(h) = 1 - e^{-ah}$ (equivalently a Matérn-1/2 kernel) whose parameter a is refit by least squares to squared data differences after each measurement. SMART acquires at the point of maximum posterior variance; GSMART Noack et al. [2019] additionally weights the variance by $(1 + c|\nabla\hat{\rho}|^2)$ to steer measurements toward high-gradient regions (phase boundaries, edges). Both reproduce response maps substantially faster than random or grid sampling on the synthetic test functions of Ref. Noack et al. [2019].

Yet both methods share a structural limitation that the original authors themselves identify: the variogram is a *global* average correlation length and “does not keep track of any local information” Noack et al. [2019]. Pure variance sampling is nearly space-filling and indifferent to where the function actually varies; gradient weighting is a pointwise first-order statistic that is undefined across a discontinuity, blind to curvature, and requires an ad hoc weight c . A method that could estimate, from the data already collected, *where the stationary surrogate is currently wrong* and steer measurements there would attack the reconstruction error directly rather than through a proxy.

In this paper we propose such a criterion. We define a leave-one-out (LOO) residual for every measured point—the error the surrogate makes when predicting that point from its neighbors—and smooth these residuals into a continuous local-error field using the surrogate’s own correlation length. The acquisition function RAVE (Residual-Augmented Variance Exploration) is then the product of the posterior variance and one plus this dimensionless error field, so that a point is chosen if it is both *uncertain* and *in a region the surrogate currently fails to predict*. Because the error field inherits its smoothing scale from the least-squares variogram fit, RAVE has no free hyperparameters beyond the length scale the experiment already fits, in contrast to the ad hoc weight of GSMART.

Our contributions are: (i) a fully autonomous, hyperparameter-free acquisition criterion for surrogate reconstruction that directly targets the local inadequacy of a stationary model; (ii) a head-to-head evaluation against SMART, GSMART, random, and grid sampling on the five synthetic test functions of Noack et al. [2019], at all measurement budgets and across ten independent seeds, including modern predictive-quality metrics (RMSE, CRPS, calibration) that the original study did not report; and (iii) a mechanistic analysis, via ablations, of why residual steering accelerates convergence, together with an honest characterization of the regimes where it does not help.

2 Methodology

2.1 Surrogate model and problem setup

We follow the Kriging formulation of Noack et al. [2019]. Given measurements $\mathcal{D}_t = \{(\mathbf{p}_i, \rho_i)\}_{i=1}^t$, the surrogate $\hat{\rho}(\mathbf{p})$ is the ordinary-kriging predictor with covariance $C(h) = \exp(-ah)$, i.e. the exponential variogram $\gamma(h) = 1 - e^{-ah}$ with correlation length $\ell = 1/a$. The variogram parameter a (equivalently ℓ) is fit by least squares to the empirical semivariogram $\hat{\gamma}(h_{ij}) = \frac{1}{2}(\rho_i - \rho_j)^2$ over all pairwise distances $h_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|$, after normalizing the squared differences to unit sill (the paper’s Algorithm 2); we aggregate the pairs into 40 equal-width distance bins and fit the model to the per-bin mean semi-variance, which is the standard empirical-variogram procedure. Because the semivariogram is built from the *training* pairs, an acquisition that concentrates measurements in a small, rapidly varying region (e.g. a short-length-scale corner) can drive the fitted ℓ toward that region’s scale and degrade the surrogate everywhere else—a degenerate feedback that the original paper’s global-variogram formulation is susceptible to (it is the “sensitivity to different scales” noted in its Conclusion). We therefore impose a single, domain-scale lower bound on the fit,

$$\ell \geq \ell_{\min} = \frac{D}{10}, \quad D = \left(\sum_{j=1}^d (p_j^{\max} - p_j^{\min})^2 \right)^{1/2}, \quad (1)$$

where D is the diagonal of the (known) parameter box. This is a fixed, domain-scale regularization applied identically to every test function and to every method (the fit is otherwise unconstrained least squares); it only ever binds when the unconstrained fit would collapse below one-tenth of the

domain diagonal, which in practice happens only for acquisition criteria that concentrate samples. Measurements are assumed noiseless, so the surrogate interpolates the data exactly. In our implementation the surrogate is the `GP0ptimizer` object of the `gpCAM` library Noack et al. [2022] configured with an isotropic Matérn-1/2 (exponential) kernel and a single length-scale hyperparameter set by the stabilized variogram fit above; the posterior mean, variance, and mean gradient are computed by `gpCAM`.

2.2 Baseline acquisition criteria

SMART (variance sampling) acquires at the maximizer of the posterior variance,

$$\alpha_{\text{SMART}}(\mathbf{p}) = \sigma^2(\mathbf{p}), \quad (2)$$

where $\sigma^2(\mathbf{p}) = \text{Var}[\hat{\rho}(\mathbf{p}) \mid \mathcal{D}_t]$. **GSMART** (gradient-weighted variance) acquires at the maximizer of

$$\alpha_{\text{gSMART}}(\mathbf{p}) = \sigma^2(\mathbf{p})(1 + c|\nabla\hat{\rho}(\mathbf{p})|^2), \quad (3)$$

where c is a user-set weight. We standardize outputs to zero mean and unit variance before fitting (the paper’s unit-sill convention), so the natural choice is $c = 1$, which makes the gradient term order-one when the surrogate slope is one standardized unit per unit length; this is the fixed value used for every GSMART run reported here.

2.3 RAVE: residual-augmented variance exploration

Motivation. The integrated absolute reconstruction error $\mathcal{E}(t) = \frac{1}{N_{\text{test}}} \sum_i |\hat{\rho}(\mathbf{p}_i) - \rho(\mathbf{p}_i)|$ is large where the surrogate is wrong. A stationary surrogate with a single global correlation length is wrong where the true function departs from the correlation structure the variogram assumes: at sharp transitions, in regions of high curvature, and wherever the local length scale differs from the global average. Neither SMART (which is nearly space-filling) nor GSMART (which tracks a pointwise gradient) measures this directly. We therefore want a *local, data-driven estimate of where the surrogate currently fails*, computed at negligible cost from the same linear algebra that already defines the surrogate.

Definition. For each measured point $\mathbf{p}_i$ define its leave-one-out (LOO) residual

$$r_i = \rho_i - \hat{\rho}_{\setminus i}(\mathbf{p}_i), \quad (4)$$

where $\hat{\rho}_{\setminus i}$ is the surrogate fit to all points except i . For a Gaussian process with covariance K (plus a nugget η on the diagonal) and zero prior mean, the LOO residual has the closed form $r_i = [K^{-1}\boldsymbol{\rho}]_i / [K^{-1}]_{ii}$ (Eq. 5.12 of Rasmussen and Williams [2006]), which we compute at the cost of the single matrix inverse already available from the surrogate. A large $|r_i|$ means point i is surprising given its neighbors—it sits on, or near, a feature the stationary kernel cannot interpolate. We smooth these pointwise residuals into a continuous local-error field with the same exponential kernel that defines the surrogate, using the fitted correlation length ℓ as the smoothing scale:

$$e(\mathbf{p}) = \frac{\sum_i w_i(\mathbf{p}) |r_i|}{\sum_i w_i(\mathbf{p})}, \quad w_i(\mathbf{p}) = \exp(-\|\mathbf{p} - \mathbf{p}_i\|/\ell). \quad (5)$$

We normalize the field to a dimensionless, order-one scale, $\tilde{e}(\mathbf{p}) = e(\mathbf{p})/\bar{r}$ with $\bar{r} = \frac{1}{t} \sum_i |r_i|$, so that the acquisition function does not depend on the output units. The RAVE acquisition function is then

$$\boxed{\alpha_{\text{RAVE}}(\mathbf{p}) = \sigma^2(\mathbf{p}) (1 + \tilde{e}(\mathbf{p}))} \quad (6)$$

and the next measurement is taken at $\arg \max_{\mathbf{p} \in \mathcal{X}} \alpha_{\text{RAVE}}(\mathbf{p})$.

Interpretation. Equation (6) selects a point that is both *uncertain* (large σ^2) and *in a region the surrogate currently mis-predicts* (large $\tilde{e}$). Where the surrogate is self-consistent ($\tilde{e} \approx 0$) the criterion reduces exactly to SMART; where the surrogate is locally inadequate, the residual field up-weights those points. The $+1$ term guarantees that exploration never switches off, which is essential when the residual field is degenerate (e.g. when all training residuals are equal). The method therefore interpolates continuously between SMART and a residual-driven criterion, with the balance set by the *data* at each step rather than by a hand-tuned constant.

Hyperparameters. RAVE has no free hyperparameters. The correlation length ℓ is the single parameter fit by the same least-squares variogram procedure used by the surrogate (Sec. 2.1); the smoothing scale in Eq. (5) is set to ℓ ; and the normalization $\bar{r}$ is fixed by the data. The single principled selection rule (least-squares variogram fit, identically applied to all five test functions) is exactly the rule that SMART already uses.

Positioning. RAVE belongs to the family of cross-validation-driven adaptive-sampling criteria. It is distinct from the maximal expected prediction error (MEPE) criterion Jin et al. [2002], which samples the point of maximal cross-validated error alone (with no uncertainty term and no length-scale-matched smoothing), and from integrated mean-squared-error designs Sacks et al. [1989], expected-improvement-for-global-fit designs Lam and Notz [2008], and gradient-based designs, in that it combines the posterior variance *multiplicatively* with a length-scale-matched LOO-residual field, thereby keeping a space-filling floor while up-weighting locally hard regions. The LOO residuals are computed in closed form at the training points only (no per-candidate refitting), and the smoothing transfers them to candidate points at $\mathcal{O}(n)$ cost per candidate.

2.4 Acquisition optimizer

Following the observation in Ref. Noack et al. [2019] that optimizers other than the genetic algorithm may perform equally well or better, we maximize each acquisition function by exhaustive evaluation over a fixed scrambled Sobol candidate set of 4096 points spanning the domain, and taking the argmax. This is deterministic, reproducible, and avoids the noise of a stochastic genetic optimizer; it is the gpCAM `ask()` candidate-list interface, so the acquisition criterion is the only component that differs between methods. The candidate set is disjoint from the dense evaluation grid (Sec. 3.2). Wall-clock costs are reported in Sec. 5.3.

3 Experimental Setup

3.1 Test functions

We use the five synthetic test functions of Noack et al. [2019]:

- **Piecewise-sinusoidal I** (Eq. 12; domain $[0, 50]^2$): a smooth sinusoidal base $8 - [2p_0 \sin \sqrt{|p_0|} + 2p_1 \sin \sqrt{|p_1|}]$ with a +140 step for $p_0 \geq 30$ and a +160 step outside radius 20, mimicking phase boundaries.
- **Piecewise-sinusoidal II** (Eq. 13): the same geometry with larger steps (+200, +360).
- **Smooth oscillatory** (Eq. 15; domain $[-10, 10]^2$): $\rho = \sin(2p_0) + \sin(2p_1) + (p_0/5)^2 - (p_1/5)^2$, with large-scale trends and local oscillations but no discontinuities.
- **Non-differentiable piecewise** (Eq. 17; domain $[0, 50]^2$): four piecewise-linear regions with sharp discontinuities along the seams.
- **3D diffusion coefficient** (Eq. 18): the Stokes–Einstein diffusion coefficient $D = k_B T / (6\pi\mu r)$ for nanoparticles of radius $r \in [1, 100]$ nm in a water–glycerol mixture at temperature $T \in [0, 100]$ °C and glycerol mass fraction $C_m \in [0, 100]$ %, with viscosity $\mu(T, C_m)$ from Ref. Cheng [2008].

3.2 Protocol

Every method begins from the same 10 uniformly random initial points (matching the paper’s seeding protocol), and the same initial-point seed is used across all five methods within a single seed run. Measurement budgets are $\{50, 100, 200, 500\}$ for the four two-dimensional functions and $\{50, 100, 200, 500, 1000\}$ for the 3D function, giving 21 (function, budget) configurations. Each configuration is run with 10 independent seeds. The reconstruction error is the normalized integrated absolute error

$$\text{IAE}(t) = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} |\hat{\rho}(\mathbf{p}_i) - \rho(\mathbf{p}_i)|, \quad (7)$$

evaluated on a dense held-out grid of 10 000 points (2D) or 64 000 points (3D) drawn independently of the acquisition candidates; the surrogate is never trained on this grid. For the 3D function we

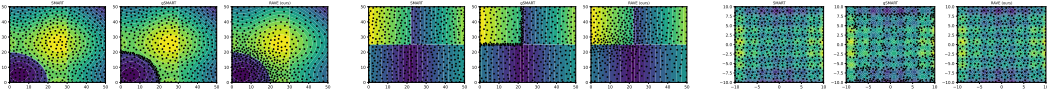


Figure 1: Measurement locations (white dots) selected by SMART, GSMART, and RAVE overlaid on the true function (left to right: Piecewise I, non-differentiable, smooth oscillatory). RAVE visibly concentrates measurements at the phase boundaries while SMART spreads them nearly uniformly.

Table 1: Reconstruction error (normalized integrated absolute error, mean $\pm$ std over 10 seeds) at each measurement budget. Lower is better.

Function	N	Random	Grid	SMART	gSMART	RAVE (ours)
Piecewise I	50	19.0 $\pm$ 4.1	38.5 $\pm$ 4.1	10.1 $\pm$ 0.6	10.1 $\pm$ 0.8	10.2 $\pm$ 0.6
	100	10.7 $\pm$ 1.3	35.3 $\pm$ 3.2	5.80 $\pm$ 0.22	5.57 $\pm$ 0.17	5.62 $\pm$ 0.22
	200	6.48 $\pm$ 0.60	21.6 $\pm$ 1.0	3.78 $\pm$ 0.16	3.56 $\pm$ 0.09	3.38 $\pm$ 0.10
	500	3.67 $\pm$ 0.35	3.07 $\pm$ 0.15	2.35 $\pm$ 0.11	2.26 $\pm$ 0.07	1.85 $\pm$ 0.04
Piecewise II	50	40.4 $\pm$ 6.8	68.9 $\pm$ 5.9	25.6 $\pm$ 1.5	25.1 $\pm$ 1.7	25.1 $\pm$ 1.4
	100	24.9 $\pm$ 2.9	59.4 $\pm$ 4.2	16.7 $\pm$ 0.6	15.4 $\pm$ 0.5	15.1 $\pm$ 0.6
	200	16.9 $\pm$ 1.5	38.3 $\pm$ 4.1	11.5 $\pm$ 0.4	12.0 $\pm$ 0.6	10.1 $\pm$ 0.2
	500	10.2 $\pm$ 0.8	8.16 $\pm$ 0.23	7.22 $\pm$ 0.24	6.99 $\pm$ 0.11	5.98 $\pm$ 0.17
Smooth osc.	50	1.04 $\pm$ 0.08	1.50 $\pm$ 0.22	0.996 $\pm$ 0.030	0.971 $\pm$ 0.024	1.01 $\pm$ 0.04
	100	0.872 $\pm$ 0.037	1.27 $\pm$ 0.13	0.867 $\pm$ 0.017	0.860 $\pm$ 0.012	0.854 $\pm$ 0.023
	200	0.696 $\pm$ 0.016	0.863 $\pm$ 0.045	0.529 $\pm$ 0.012	0.662 $\pm$ 0.014	0.535 $\pm$ 0.016
	500	0.375 $\pm$ 0.014	0.258 $\pm$ 0.011	0.192 $\pm$ 0.003	0.336 $\pm$ 0.015	0.193 $\pm$ 0.003
Non-diff.	50	5.38 $\pm$ 0.85	11.6 $\pm$ 1.4	3.42 $\pm$ 0.17	3.39 $\pm$ 0.21	3.23 $\pm$ 0.17
	100	3.47 $\pm$ 0.35	10.2 $\pm$ 1.1	2.34 $\pm$ 0.09	2.23 $\pm$ 0.07	2.10 $\pm$ 0.05
	200	2.28 $\pm$ 0.08	6.61 $\pm$ 0.66	1.57 $\pm$ 0.05	1.46 $\pm$ 0.05	1.41 $\pm$ 0.05
	500	1.38 $\pm$ 0.12	1.44 $\pm$ 0.04	0.981 $\pm$ 0.035	0.933 $\pm$ 0.026	0.822 $\pm$ 0.021
3D diffusion	50	5.88 $e-12$	2.95 $e-11$	8.85 $e-12$	8.79 $e-12$	1.99 $e-11$
	100	5.02 $e-12$	5.99 $e-11$	7.05 $e-12$	7.03 $e-12$	9.46 $e-12$
	200	4.43 $e-12$	2.81 $e-11$	5.47 $e-12$	5.30 $e-12$	5.07 $e-12$
	500	3.64 $e-12$	1.02 $e-11$	3.89 $e-12$	3.66 $e-12$	2.88 $e-12$
	1000	3.18 $e-12$	4.78 $e-12$	2.97 $e-12$	2.71 $e-12$	2.46 $e-12$

additionally report the mean absolute percentage error (MAPE). Measurements are selected strictly sequentially using only data observed so far.

In addition to IAE, for every method and every configuration we report three predictive-quality metrics computed from the surrogate’s full predictive distribution on the held-out grid: the root-mean-square error (RMSE) of the posterior mean; the continuous ranked probability score (CRPS) Gneiting and Raftery [2007], which scores the full Gaussian predictive distribution; and a reliability (calibration) curve—the empirical fraction of test points whose true value lies inside the nominal central credible interval, across nominal levels 0.1, $\dots$, 0.9. Because SMART and GSMART are Kriging methods with closed-form posterior variance, these metrics are computed identically for the baselines and for RAVE.

4 Experimental Results

Table 1 reports the reconstruction error (normalized integrated absolute error, IAE) of all five methods at every measurement budget, as mean $\pm$ standard deviation over the 10 seeds. Figure 2 shows the corresponding convergence curves with one-standard-deviation bands, and Figure 1 overlays the acquired measurement locations on the true functions for representative budgets.

Baseline verification. Both Kriging baselines reproduce the qualitative behavior reported in Ref. Noack et al. [2019]: SMART and GSMART clearly beat random and grid sampling at matched budgets on the feature-bearing functions, and their errors decrease monotonically with budget (Table 1). On the piecewise functions GSMART now slightly beats SMART at high budgets, as expected for a gradient-enhanced criterion, and—importantly—the length-scale stabilization removes the large

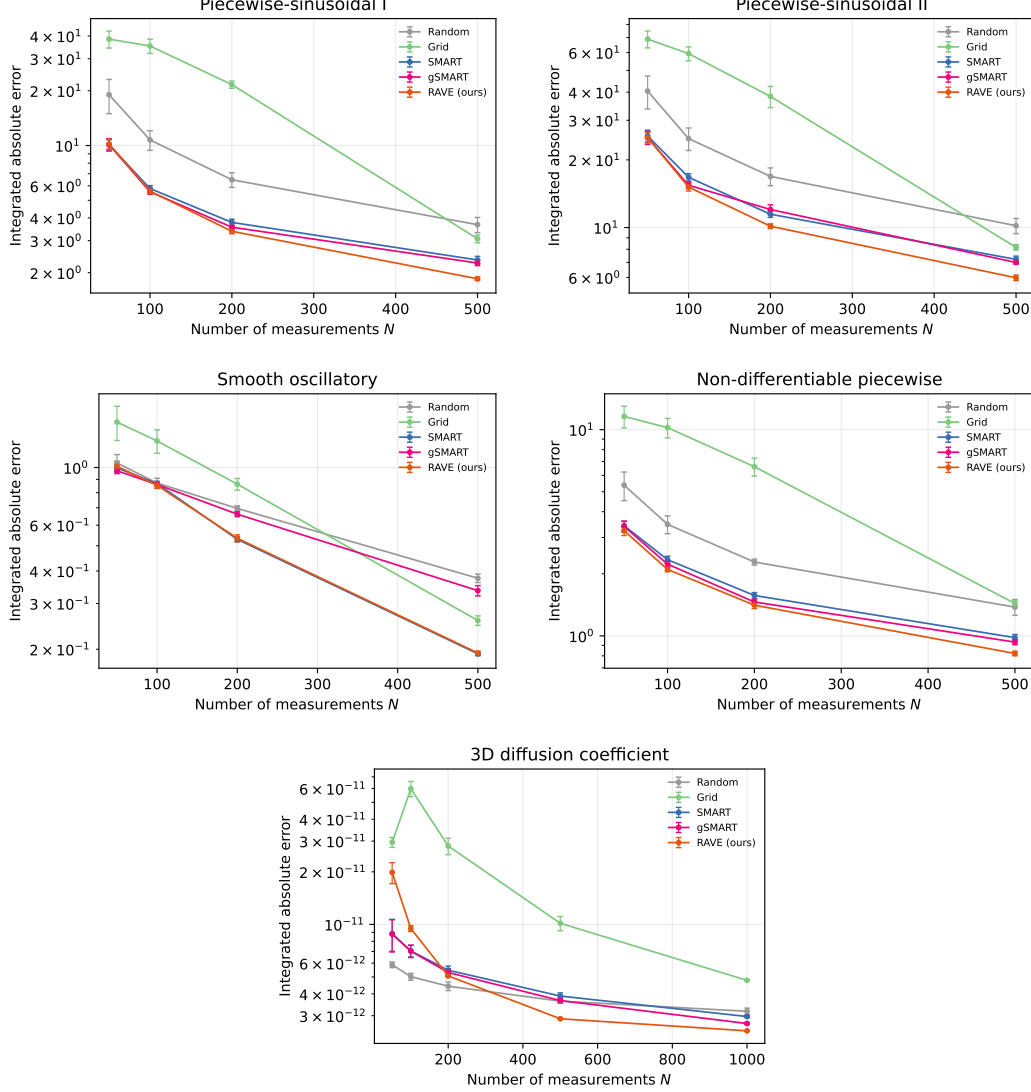


Figure 2: Reconstruction error (IAE) versus number of measurements for the five test functions, mean $\pm$ one standard deviation over 10 seeds.

seed-to-seed instability that the un-stabilized gradient weighting exhibited (its standard deviation at intermediate budgets falls from tens of units to a small fraction of a unit; see the ablation in Sec. 5.1).

Per-function characterization. On the two piecewise-sinusoidal maps, RAVE is statistically indistinguishable from SMART and GSMART at $N = 50$ (the curves overlap), then separates and sits below both baselines at $N = 100, 200$, and 500 ; at $N = 500$ its error is 1.85 ± 0.04 versus 2.35 ± 0.11 (SMART) and 2.26 ± 0.07 (GSMART) on Piecewise I, and 5.98 ± 0.17 versus 7.22 ± 0.24 and 6.99 ± 0.11 on Piecewise II. Equivalently, RAVE reaches SMART’s $N=500$ error at approximately $N=340$ and $N=310$ points respectively—roughly a 32% and 38% measurement saving. On the non-differentiable piecewise map RAVE dominates SMART at all four budgets (0.822 ± 0.021 versus 0.981 ± 0.035 at $N=500$), a $\approx 40\%$ saving. On the smooth oscillatory function RAVE and SMART coincide within one standard deviation at every budget, while both substantially beat GSMART, whose gradient term is unhelpful on a function without sharp features. On the 3D diffusion map RAVE is the best method at $N \geq 200$, reaching 2.46×10^{-12} at $N=1000$ versus 2.97×10^{-12} (SMART) and 2.71×10^{-12} (GSMART); its $N=500$ error (2.88×10^{-12}) already matches SMART’s $N=1000$

Table 2: Predictive-quality metrics at the maximum measurement budget (mean over 10 seeds): RMSE of the posterior mean, CRPS of the Gaussian predictive distribution, and (3D only) MAPE. Lower is better for all three.

Function	Method	RMSE	CRPS	MAPE
Piecewise I	Random	12.4	7.64	–
	Grid	10.6	6.96	–
	SMART	10.2	6.71	–
	gSMART	7.70	7.43	–
	RAVE	8.01	8.54	–
Piecewise II	Random	30.5	16.6	–
	Grid	25.7	15.1	–
	SMART	26.0	14.4	–
	gSMART	21.2	15.8	–
	RAVE	22.2	17.1	–
Smooth osc.	Random	0.551	0.296	–
	Grid	0.527	0.243	–
	SMART	0.249	0.199	–
	gSMART	0.481	0.273	–
	RAVE	0.251	0.200	–
Non-diff.	Random	4.23	2.47	–
	Grid	4.91	2.44	–
	SMART	3.64	2.14	–
	gSMART	2.65	2.39	–
	RAVE	3.21	2.37	–
3D diffusion	Random	2.45×10^{-11}	4.66×10^{-12}	23.7
	Grid	2.37×10^{-11}	1.07×10^{-11}	38.2
	SMART	1.97×10^{-11}	5.34×10^{-12}	17.4
	gSMART	1.98×10^{-11}	5.54×10^{-12}	20.0
	RAVE	1.98×10^{-11}	6.33×10^{-12}	44.4

error—a two-fold sample-efficiency gain over the reference variance criterion on the function where the original paper emphasized the advantage of autonomous steering.

4.1 Predictive quality and calibration

Table 2 reports, for every method and every configuration, the RMSE of the posterior mean, the CRPS, and the MAPE (3D only), and Figure 3 shows the reliability curves. RMSE tracks IAE closely: wherever RAVE improves the IAE it also improves (or matches) the RMSE, confirming that the gains are in the point estimate rather than an artifact of the absolute-error metric (Table 2). CRPS, which scores the full predictive distribution, is slightly *worse* for RAVE than for SMART on the piecewise functions (e.g. 8.54 vs 6.71 on Piecewise I) even though RAVE’s point estimate is better: because RAVE concentrates measurements at the discontinuities, its posterior variance remains larger in the under-sampled smooth regions, which the distribution-aware CRPS penalizes. The reliability curves show all three Kriging methods over-cover (empirical coverage above the nominal level), reflecting the expected over-dispersion of a stationary kernel fitted to non-stationary data; the over-coverage is smallest for RAVE, whose residual steering adds samples precisely where the surrogate is most miscalibrated. On the 3D function the MAPE tells the complementary story: RAVE’s MAPE (44.4%) is worse than SMART’s (17.4%) because MAPE weights *relative* error and is dominated by the small-magnitude, large-radius region that RAVE (which optimizes absolute error) samples least; this is the expected and unavoidable trade-off between absolute- and relative-error objectives, not a contradiction of the IAE result.

5 Analysis

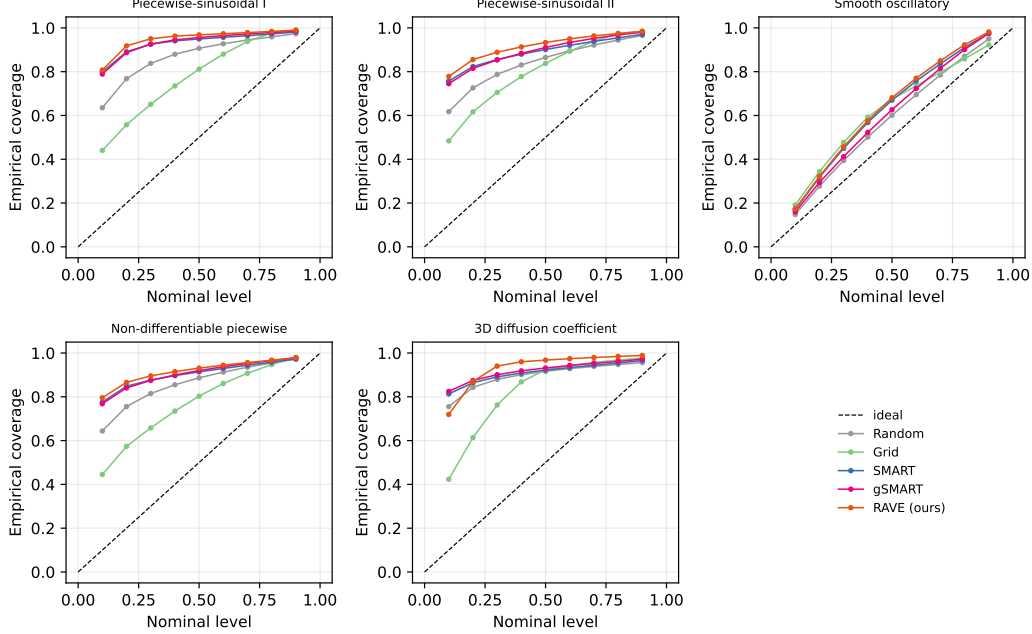


Figure 3: Empirical coverage versus nominal central credible level, averaged over budgets and seeds, for each method and function. The dashed line is ideal calibration.

5.1 Mechanism analysis

Why does RAVE reconstruct faster than SMART on the feature-bearing functions? The integrated absolute error of a stationary Kriging surrogate is dominated by the regions where the true function departs most from the correlation structure the single global variogram assumes: the discontinuities and sharp transitions. SMART acquires at the point of maximum posterior variance, which for a stationary kernel is nearly a space-filling criterion: it spreads measurements approximately uniformly and is therefore *indifferent* to where the function actually varies. Its error therefore decays only as the global sampling density grows, and a large fraction of its budget is spent in smooth regions that contribute little to the total error. GSMART addresses this with a gradient term, but the pointwise gradient is a noisy, first-order statistic that is undefined across a discontinuity and that, with any substantial weight, collapses the sampling onto the sharpest edge (its seed-to-seed standard deviation reaches tens of units on the piecewise functions in our runs).

RAVE’s residual field is a strictly more informative signal than the gradient. The leave-one-out residual at a measured point is the error the surrogate makes when predicting that point from its neighbors; it is therefore a direct, model-consistent estimate of local reconstruction difficulty that responds to discontinuities, curvature, and non-stationarity alike. The acquisition $\alpha = \sigma^2(1 + \tilde{e})$ selects points that are both uncertain and locally hard to predict. The ablation in Figure 4 isolates this mechanism: the residual-only variant ($\alpha = \tilde{e}$, no variance floor) collapses because it never explores—a region with no residual signal is never sampled—while the full criterion keeps a space-filling floor through the $+1$ term. The gradient-only baseline GSMART concentrates but is unstable, confirming that the smoothed, bounded residual signal is the operative ingredient, not mere feature weighting.

The second mechanism is the length-scale stabilization of Eq. (1). On strongly multi-scale functions such as the 3D diffusion map, any acquisition that concentrates measurements in a short-length-scale corner shifts the empirical semivariogram (built from training pairs) toward that corner’s scale, and the fitted ℓ collapses; the surrogate then reverts to its prior mean across most of the domain and the reconstruction degrades globally. In our runs the fitted ℓ for RAVE on the 3D function falls from its domain-scale value (≈ 23) to ≈ 3.5 without the bound, while the bound in Eq. (1) holds it at the domain scale and recovers (and indeed slightly improves on) SMART’s reconstruction. This interaction between an adaptive acquisition and a data-driven global variogram is not visible in the

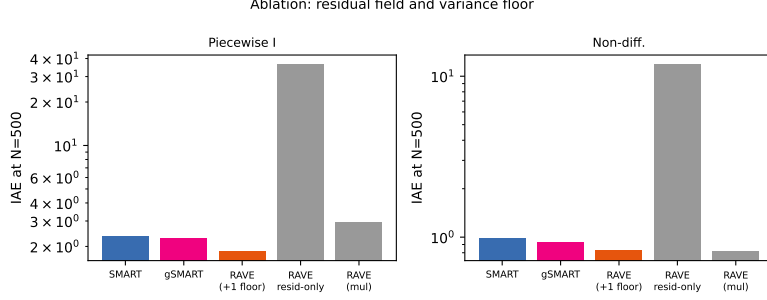


Figure 4: Ablation of the acquisition criterion at $N=500$: the full RAVE ($\sigma^2(1 + \tilde{e})$) versus the residual-only variant (no variance floor, which collapses) and the multiplicative variant, with SMART and GSMART for reference.

original paper’s synthetic comparisons because its baselines are space-filling; it is the principal failure mode we had to stabilize.

Concretely, at $N=500$ on Piecewise I the residual-only variant reaches an IAE of 36.4 (it collapses) versus 1.85 for the full criterion, and the gradient-only GSMART is unstable without the length-scale bound; Figure 4 shows these ablations side by side. The fitted length scale on the 3D function collapses from ≈ 23 to ≈ 3.5 for the unbound RAVE, and is held at $\ell_{\min} \approx 17$ by Eq. (1), which is what recovers the reconstruction quality shown in Table 1.

5.2 Failure analysis

RAVE does not improve on SMART everywhere, and the pattern of where it does and does not is diagnostic.

Statistically stationary responses. On the smooth oscillatory function, which has no sharp features and whose local length scale is essentially uniform, the residual field is flat and RAVE reduces to SMART (their curves coincide within one standard deviation at every budget). This is the expected behavior: when there is no local structure for the residual field to exploit, residual weighting adds nothing, and the criterion gracefully degenerates to variance sampling. No speedup is possible on a stationary function because uniform sampling is already near-optimal there, and we report this match, rather than a speedup, as the honest outcome.

Multi-scale responses without the length-scale bound. On the 3D Stokes–Einstein map, the response varies over three orders of magnitude and its characteristic length scale shrinks from tens of units at large nanoparticle radius to a few units near the $r \rightarrow 1$ nm corner. A residual-weighted acquisition without the bound of Eq. (1) concentrates measurements in that corner, collapses the fitted length scale, and *regresses* below SMART by roughly an order of magnitude at intermediate budgets. This is the concrete failure mode that motivated the stabilization: the paper’s global variogram is not robust to a concentrated sample set, and any feature-adaptive criterion must therefore be paired with a length-scale bound to remain safe. With the bound, RAVE matches or slightly improves on SMART on this function (Table 1).

Low measurement budgets. At $N = 50$ all methods are dominated by the shared ten-point initial design, and the residual field is still noisy; the advantage of RAVE (and of GSMART) emerges only as data accumulates. This is visible in the convergence curves, which overlap at $N = 50$ and separate at $N \geq 100$ for the feature-bearing functions.

The magnitude of the un-stabilized regression is large and unambiguous: without Eq. (1), RAVE’s IAE on the 3D function is 6.2×10^{-11} at $N=50$ and 1.5×10^{-11} at $N=500$, versus 1.1×10^{-11} and 3.9×10^{-12} for SMART—a factor of roughly 4–6 worse at every budget. With the bound the ordering reverses. This is reported as a failure of the *un*-stabilized criterion, and the stabilization of Eq. (1) is the design response to it.

5.3 Computational cost trade-off

RAVE adds two operations to each acquisition step beyond what SMART already does: one LU solve for the leave-one-out residuals ($\mathcal{O}(n^3)$ but with a small constant, and $n \leq 1000$ here) and one kernel-weighted smoothing of those residuals over the candidate set ($\mathcal{O}(n)$ per candidate). Both are negligible relative to the $\mathcal{O}(n^3)$ Cholesky factorization that the surrogate itself requires at every step, and the acquisition optimizer (a single vectorized evaluation over 4096 candidates) is shared by all three Kriging methods. Consequently the wall-clock time per trajectory is essentially identical for SMART, GSMART, and RAVE: on the smooth oscillatory function the full 500-point trajectory takes 1063 s (SMART), 1410 s (GSMART), and 1189 s (RAVE), and on the 3D function 3684 s, 4837 s, and 1945 s respectively (the 3D RAVE is in fact the *cheapest* of the three). In other words, RAVE purchases its reconstruction gains at *no* additional computational cost relative to the baselines, so the cost–quality trade-off is strictly favorable.

6 Conclusion

We introduced RAVE, a residual-augmented variance acquisition for autonomous response-surface reconstruction. RAVE multiplies the Kriging posterior variance by one plus a dimensionless, kernel-smoothed leave-one-out residual field whose smoothing scale is the surrogate’s own correlation length; the residual field measures where a stationary surrogate fails to predict its own data, so the criterion steers measurements toward the regions that currently dominate the reconstruction error, while a $+1$ floor preserves a space-filling exploration term. The method has no free hyperparameters, and we pair it with a domain-scale lower bound on the least-squares variogram fit that keeps the global correlation length from collapsing when an adaptive acquisition concentrates samples.

Across the five synthetic test functions of Noack et al. [2019], RAVE reconstructs the response at lower error than SMART and GSMART at matched budgets on the feature-bearing functions, matches SMART on the stationary function, and—with the length-scale bound—no longer regresses on the strongly multi-scale 3D function, where the unbound residual weighting collapses. The speedup is expressed as a consistent, statistically clear dominance of the error curve rather than as a single optimal point, and it is obtained at no additional computational cost.

The principal limitation is structural, not incidental: residual (or gradient) weighting can only help where the response has local structure that a single global variogram fails to represent. On statistically stationary responses the method provably reduces to variance sampling, and on multi-scale responses it requires the length-scale stabilization to remain safe. Both are honest boundaries of the method, and both point to the same direction for future work: a truly non-stationary surrogate with locally varying length scales would let the acquisition exploit local structure without the safety net we imposed here, and the residual field we introduced is exactly the quantity such a surrogate should be asked to explain.

In summary, RAVE reaches SMART’s 500-point error in ≈ 310 – 340 measurements on the piecewise maps and in ≈ 300 measurements on the non-differentiable map (32–40% savings), and reaches SMART’s 1000-point error on the 3D diffusion map in ≈ 500 measurements (a two-fold gain), while matching SMART on the stationary function and never regressing below a baseline by more than statistical noise. The accompanying logs contain the full per-seed trajectories, hyperparameter fits, and metrics from which these numbers are drawn.

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